

AI for Oceans and Robotics

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Packet Use

This printable lesson packet is generated from the paired Markdown guide. The Markdown version remains the clickable, neurodivergent-friendly working copy; this PDF carries the same lesson substance in a formal handout format. Evidence claims in this packet are based on transcript-derived lesson notes and the cleaned transcript files in this workspace.

Quick Scan

- **Grade:** Grade 2
- **Grade Hub:** Notes [`../grade-2-Notes.md`]
- **Schedule Reference:** `Spring2026_TeachingSchedule_Derek.md` [`../school/Spring2026_TeachingSchedule_I`]
- **Workbook Sheet:** 2nd in `Teaching_Placement-IGNITE Curriculum_ GCSD25-26.xlsx`
- **Lesson Focus:** Students explore how machine learning systems sort information, then connect classification ideas to early robotics and real-world problem solving.
- **CS Alignment Strength:** Strong CS lesson
- **LaTeX Source:** `ai-for-oceans-and-robotics.tex` [`../ai-for-oceans-and-robotics.tex`]
- **Bib File:** `ai-for-oceans-and-robotics.bib` [`../ai-for-oceans-and-robotics.bib`]
- **Artifacts Folder:** `artifacts/` [`../artifacts/`]

Lesson Identity

- **Likely Class Blocks:** Day 1 – 10:45-11:35 – Callon, Day 1 – 2:15-3:05 – Hennen, Day 4 – 10:45-11:35 – Barthelman
- **Main Lesson Family:** AI for Oceans and Robotics
- **Primary Evidence Base:** Transcript-derived notes plus source transcripts from this teaching-placement workspace.
- **Primary External Source Type:** Official curriculum/provider materials.

What We Are Teaching

- what AI and machine learning mean in age-appropriate language
- how a system can be trained by categories and examples
- how sorting rules affect outcomes and mistakes
- how robotics and AI both rely on clear instructions and inputs

CS Standards Alignment

- **1A-DA-06:** Collect and present the same data in various visual formats. Why it fits here: Students sort and classify examples into categories and talk about the results.
- **1A-DA-07:** Identify and describe patterns in data visualizations. Why it fits here: Students notice why the model labels some items correctly or incorrectly.
- **1A-AP-11:** Decompose the steps needed to solve a problem into a precise sequence of instructions. Why it fits here: The teacher models a repeatable login, launch, and classification routine before students work independently.

NYS / Local Curriculum Alignment

- Aligned to the local Grade 2 IGNITE workbook sheet through introductory AI literacy and structured digital problem solving.
- Supports New York-facing digital literacy and responsible technology-use goals by helping students explain what a machine is deciding and why.
- Connects data sorting to foundational computational thinking rather than treating AI as a mystery tool.

Lesson Content and Concepts

- data categories
- prediction
- machine learning
- input/output
- classification error

Evidence / Proof of Instruction

- **Transcript-derived note:** 260211 Grade 2 Callon Ai Oceans [../transcript-derived-lessons/260211-grade-2-callon-ai-oceans.md] The lesson note shows that the class explicitly introduced AI, machine learning, and the AI for Oceans task.

- **Transcript-derived note:** 260212 Grade 2 Callon Codeorg Ai [../transcript-derived-lessons/260212-grade-2-callon-codeorg-ai.md] The follow-up note documents login support and direct teaching about how students would train the model.
- **Source transcript:** 260212 Grade 2 Callon Codeorg Ai [../transcripts/260212-grade-2-callon-codeorg-ai.md] The source transcript records the teacher saying students would “train an AI bot” and move between video, discussion, and guided activity.
- **Likely student proof:** Completed AI for Oceans tasks, correct category choices, student explanations of why an item belongs in a group, and discussion of mistakes all count as proof of instruction.

Assessment Evidence

- Student can explain AI as a tool that learns from examples.
- Student can sort at least one item correctly and explain the reason.
- Student can identify when the model or choice needs to be fixed.

UDL and Inclusion Supports

Multiple Representation

- Introduce AI with a video, teacher talk, and concrete examples from daily life.
- Use visible category labels and color coding when sorting examples.
- Keep the difference between “example” and “guess” explicit.

Multiple Action / Expression

- Allow students to answer by sorting, pointing, talking, or dragging items.
- Pair students so one can navigate while the other explains.
- Provide printed category cards for off-screen rehearsal before the digital task.

Multiple Engagement

- Connect ocean cleanup to a meaningful real-world problem.
- Alternate short discussion bursts with active sorting.
- Treat mistakes as useful training data, not failure.

How We Are Already Making It Inclusive

- The transcript shows strong whole-group framing before students were released into the task.
- Teacher modeling of login and section-code entry lowered access barriers.
- The activity already included a real-world problem context, which supports engagement for many learners.

How to Improve Inclusion Next Time

- Add picture-supported category cards for students who need more off-screen scaffolding.
- Include a bias/error conversation using one intentionally tricky example.
- Prepare sentence stems such as “I put it here because...” and “The computer guessed wrong because...”
- Save screenshots of correct and incorrect classifications in **artifacts/** as evidence examples.

Materials and Setup

- Student devices
- Code.org / AI for Oceans activity
- Projected teacher model
- Optional printed category cards

Teacher Moves / Script / Routines

- Model login and launch before discussing AI concepts.
- Stop after the first classification to ask what the computer is learning.
- Normalize wrong guesses as part of training the model.
- Close with a short reflection on where students see sorting or prediction in real life.

Common Errors and Debugging

- Students may confuse AI with a human-like robot instead of a trained model.
- Login friction can consume too much time if not tightly modeled.
- Students may sort by appearance without explaining their category logic.

Extensions and Simplifications

- Extension: compare two hard-to-sort examples and debate the best category.
- Simplification: reduce the task to fewer categories and more teacher-led sorting.
- Extension: connect the lesson to responsible AI and why bad data causes bad results.

Related Local Materials

- No direct local lesson packet linked yet.

Artifacts Checklist

- Compiled lesson PDF in **artifacts/**
- At least one screenshot, student example, or setup photo
- One short assessment or observation record
- Updated standards/source bibliography once the **.bib** file is revised
- Any teacher reflection or revision notes after reteaching

Selected Official Sources

Selected official sources that informed this packet are listed below. ocite*

References